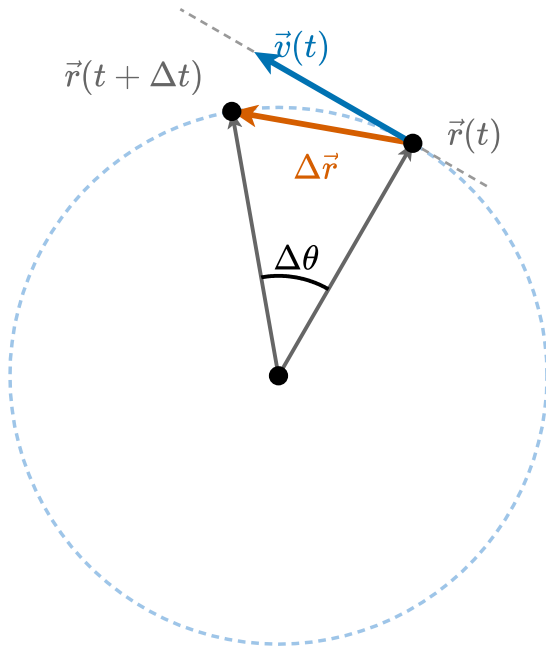


tangent line



$$\|\Delta\vec{r}\| = 2r \sin \frac{\Delta\theta}{2} \approx r \Delta\theta$$

$$\Rightarrow v = \lim_{\Delta t \rightarrow 0} \frac{\|\Delta\vec{r}\|}{\Delta t} = r \omega$$